Next-Gen Adult Incontinence Product Concept

A Microfluidic & Biomechanical Hybrid Architecture for Absolute Rewet Prevention

Document Type: Technical Concept Summary

Focus Area: Adult Incontinence / Fluidic Nonwovens

Key Innovations: Microfluidic Valves + Floating Core + 3D ADL

1. EXECUTIVE SUMMARY & CORE PARADIGM SHIFT

Traditional adult incontinence products suffer from a fundamental design flaw: they are scaled-up infant diapers. However, adult voiding mechanics, skin fragility, and body weight distribution are radically different from infant physiology. An adult urge void delivers a high volume (*250–400 mL*) at high velocity (*> 25 mL/s*), and body weight creates heavy seated compression loads (*50–80 kPa*) that crush wet cores, causing fluid rewet and severe Incontinence-Associated Dermatitis (IAD).

This document summarizes an advanced multi-layered adult brief architecture that integrates **Microfluidic One-Way Valve Arrays**, a **3D micro-channeled fluidic ADL**, **wide micro-foam perimeter gaskets**, a **floating core suspension architecture**, and a **30% fluff-reduced core matrix**. Together, these elements guarantee zero fluid rewet, zero elastic skin shearing, and instant fluid acquisition.

2. UNADDRESSED INDUSTRY SHORTFALLS ADDRESSED BY CONCEPT

A. Seated Hydrostatic Pressure ("Rewet")

Standard testing measures static capacity, but a seated 70–90 kg adult exerts immense pressure directly on the wetted core. Saturated Superabsorbent Polymer (SAP) squeezed under this load releases fluid back through the top sheet against the skin.

B. High-Velocity Intake Failure

An adult urge void overpowers slow-absorbing SAP, causing fluid to spill over leg cuffs before acquisition occurs. Traditional designs rely on thick fluff pulp to temporarily store fluid, adding unnecessary bulk.

C. Elastic Shearing & Skin Tears

Narrow elastic strings concentrate tension into tiny surface areas (> 32 mmHg), exceeding micro-vascular capillary occlusion pressure and causing skin red marks, tears, and breakdown in elderly patients.

D. Ammonia & Alkaline pH Spikes

Bacterial urease converts urine urea into free ammonia (NH_3), driving urine pH from acidic (pH ≈ 6.0) to alkaline (pH > 8.0). This destroys the skin's protective acid mantle and activates digestive enzymes.

3. STRUCTURAL INNOVATIONS & ARCHITECTURAL ELEMENTS

System Structural Layout (Cross-Section View)

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                [ User Skin Contact Layer ] (Hydrophilic Bonded Top Sheet)
                                                   │
       ├── [ Wide Micro-Foam Gaskets ] ──────── (Open-cell PU Foam, < 20 mmHg pressure, lateral dam)
                                                   │
[ Microfluidic One-Way Valve Array ] ───── (Passive elastomeric check valves: Open downward / Seal upward)
                                                   │
    [ 3D Micro-Channeled ADL ] ────────────── (Directional fluid conduits; longitudinal intake engine)
                                                   │
      [ Suspended / Floating Core ] ──────────── (30% Fluff-Reduced SAP matrix in isolated pockets)
       │ └─────── [ Air-Gap Decoupling Zone ] ─ (Tension hammock effect; isolates seated compression)
     [ Outer Waterproof Chassis ] ───────────── (Breathable microporous backsheet + structural shell)
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Component 1: Microfluidic One-Way Valve Array (Active Check-Valve Barrier)

- **Directional Fluidic Mechanics:** Embedded directly beneath the top sheet or integrated into the upper ADL surface, this layer features millions of microscopic, flexible one-way check valves (duckbill or flap micro-geometries).
- **Downward Intake vs. Upward Occlusion:** Downward fluid pressure from a void pushes the flexible micro-flaps open, allowing urine to enter the 3D channels rapidly. Under upward hydrostatic seating pressure ($50\text{–}80\ kPa$), reverse pressure forces the micro-valve flaps firmly shut, mechanically locking liquid beneath the top sheet and rendering backflow physically impossible.

Component 2: 3D Micro-Channeled Acquisition-Distribution Layer (ADL)

- **Hydraulic Routing Conduits:** Replaces bulky fluff pulp padding buffer with thermo-bonded micro-grooves that use capillary action and fluid pressure to transport urine longitudinally across the full diaper core in seconds.
- **Direct SAP Injection:** Routes high-velocity fluid directly to un-utilized core zones, preventing surface pooling and side-gushing leakage.

Component 3: Wide Micro-Foam Gaskets (Leg Seals & Lateral Dam)

- **Capillary Occlusion Protection:** Uses 15–20 mm wide strips of open-cell, breathable polyurethane foam instead of sharp elastic strings. Keeps interface pressure below $20\ mmHg$ (below the $32\ mmHg$ tissue ischemia threshold).
- **Hydrostatic Damming & Breathability:** Provides a soft perimeter dam that prevents lateral squeeze-out under seated load while allowing moisture vapor to escape, eliminating skin maceration along crease lines.

Component 4: Floating Core Suspension Architecture

- **Mechanical Decoupling ("Hammock Effect"):** The absorbent pad is attached to the outer chassis only at front and rear waistbands, allowing the sides to "float."
- **Rewet Elimination:** Body weight compresses the outer chassis against the chair seat while the body rests on the suspended inner chassis tension, shielding the saturated core from direct crushing forces.

Component 5: 30% Fluff-Reduced SAP-Optimized Core Matrix

- **Optimal Hybrid Ratio:** Retains 70% fluff pulp combined with high-retention SAP in compartmentalized pockets to provide sufficient capillary network without gel-blocking.
- **Profile Thickness Reduction:** Reduces overall diaper thickness by nearly one-third, enhancing discreetness without sacrificing total fluid capacity ($> 800\ mL$).

4. CROSS-SECTIONAL TECHNICAL SPECIFICATIONS

Layer / Component	Material Composition	Primary Function	Key Engineering Metric
1. Skin Top Sheet	Hydrophilic bicomponent nonwoven (Spunbond/Meltblown)	Instant fluid passage; ultra-soft skin contact	Strike-through time < 1.8 s
2. Microfluidic Valve Layer	3D Micro-perforated elastomeric film / Flap aperture array	One-way check valve (Passes urine down, locks rewet up)	Reverse burst pressure > 90 kPa; Zero rewet
3. Perimeter Gaskets	Open-cell Polyurethane Micro-Foam (15–20mm width)	Soft leg seal & lateral hydrostatic dam	Interface pressure < 20 mmHg
4. 3D Channeled ADL	3D embossed high-loft carded nonwoven / Conical micro-film	Directional fluid routing; surge buffer replacement	Longitudinal wicking speed > 15 cm/s
5. Floating Core	70% fluff pulp + 30% reduction; High-capacity SAP pockets	Fluid storage and locking under pressure	Retention capacity > 800 mL; 30% bulk reduction
6. Suspended Air-Gap	Decoupled air space between core and outer shell	Seating pressure isolation; rewet prevention	Seated rewet reduction > 95%
7. Outer Chassis	Breathable microporous Polyethylene film + Cloth-like outer	Leak-proof outer barrier with vapor permeability	MVTR > 2500 g/m²/24h

5. CLINICAL & BIOMECHANICAL OUTCOMES

Skin Integrity & IAD Prevention

The combination of microfluidic check valves and wide foam gaskets guarantees zero liquid wetback to the stratum corneum, completely preventing urine-induced maceration and skin shearing.

User Dignity & Ergonomics

A 30% thinner profile eliminates the bulky silhouette under normal clothing. The flexible floating core moves naturally during ambulation, preventing sag drift and groin chafing.

6. MANUFACTURING & COMMERCIAL IMPLEMENTATION

- **Microfluidic Film Lamination:** Microfluidic apertured films can be inline laser-perforated or micro-embossed at standard web converter line speeds (> 300 m/min).
- **Continuous Roll Ultrasonic Welding:** Foam gaskets and core suspension anchor points are bonded using high-speed ultrasonic spot welding, eliminating stiff glue lines that irritate fragile skin.
- **Patterned SAP Dosing:** Modern compartmentalized core dosing lines allow precise positioning of SAP in channel-aligned pockets without complete machine re-tooling.